ChronoCTI: Mining Knowledge Graph of Temporal Relations among Cyberattack Actions

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***Abstract*—Cyberthreat intelligence (CTI) reports on past cyberattacks describe the sequence of actions of attackers in terms of time. The sequence contains temporal relations among attack actions, such as *a malware is first downloaded and then executed*. Information related to temporal relations enables cybersecurity practitioners to investigate past cyberattack incidents and analyze attackers' behavior. However, cybersecurity practitioners must extract such information automatically, in a structured manner, through a common vocabulary to reduce human effort and enable sharing, and collaboration. *The goal of this paper is to aid security practitioners in proactive defense against attacks by automatic information extraction of temporal relations among attack actions from cyberthreat intelligence reports*. We propose ChronoCTI, an automated pipeline for extracting temporal relations among attack actions from CTI reports. The attack actions are represented as MITRE ATT&CK techniques, and the relations are represented as a knowledge graph. To construct ChronoCTI, we build a ground truth dataset of temporal relations and apply large language models, natural language processing, and machine learning techniques. ChronoCTI demonstrates higher precision but lower recall performance on a real-world dataset of 94 CTI reports. ChronoCTI achieves macro precision, recall, and F1 scores of 0.75, 0.46, and 0.54, respectively. ChronoCTI aids practitioners in analyzing large volumes of CTI reports, thinking like attackers, and knowing what attack actions are likely to happen next, which enables the practitioners to assess imminent threats and strengthen their cybersecurity readiness.**

***Index Terms*—Tactics, Techniques, and Procedures, MITRE ATT&CK, Temporal relation, Cyberthreat intelligence, CTI reports, Knowledge graph, attack graph, TimeML**

I. Introduction

Cybersecurity vendors publish technical reports on cyberattack incidents, commonly known as cyberthreat intelligence (CTI) reports. The reports describe actions executed by attackers, what tools were used, and how the defense was bypassed. Cybersecurity practitioners use CTI reports as a source of information for investigating past cyberattacks and analyzing attackers' behaviors. Effective and efficient information extraction from CTI reports improves defenders' ability to assess the threats and take defensive actions.

CTI reports contain information regarding temporal relations among attack actions, which refers to the sequence of attack actions as time progresses. Fig. 1 shows an excerpt from a CTI report, where we can see that *phishing* action is followed by *process injection* action in terms of time. The example indicates a temporal relation between two actions: *phishing* happens before *process injection*. Effective and efficient extraction of such temporal information requires the following: (a) the extraction should be automated to minimize the need for human intervention; (b) the extracted information should be structured to enable sharing; and (c) a common vocabulary should represent the extracted information to enable collaboration and universal understanding among researchers, and practitioners. Thus, information extraction of temporal relation among attack actions from CTI reports can benefit practitioners by (a) reducing human effort and time to parse complex and large volumes of CTI reports and (b) providing proactive CTI on how adversaries operate by composing multiple actions. *The goal of this paper is to aid security practitioners in proactive defense against attacks by automatic information extraction of temporal relations among attack actions from cyberthreat intelligence reports*. We address the research question (RQ): How do we automatically extract information on temporal relations among attack actions from cyberthreat intelligence reports, where the information is structured and uses a common vocabulary?

To address the RQ, we propose **ChronoCTI**, which extracts temporal relations among attack actions from CTI reports. ChronoCTI automatically extracts information by using natural language processing and machine learning techniques. To represent the extracted information in a structured manner, ChronoCTI outputs information as knowledge graphs. Finally, ChronoCTI uses MITRE ATT&CK, which is the most comprehensive taxonomy of attack actions, to represent the information in a common vocabulary. Fig. 1 shows the capability of ChronoCTI, where a knowledge graph of temporal relations among MITRE ATT&CK techniques is extracted from a passage of a CTI report. To our knowledge, ChronoCTI is the first end-to-end pipeline for automatically mining the knowledge graph, specifying what attack actions happened one after another. We open-source the dataset and machine learning models[1] for use, modification, and further enhancements.

II. Related work

The section discusses several studies related to our work.

Temporal extraction Literature contains studies on extracting (a) temporal information from text, such as date and time, and (b) temporal relations between events in texts, such as *cooking* before *eating*. Several datasets and specifications exist for temporal extraction tasks: (a) TimeBank [2], a corpus of

[1]https://figshare.com/s/0e7bda617c9f4a668655

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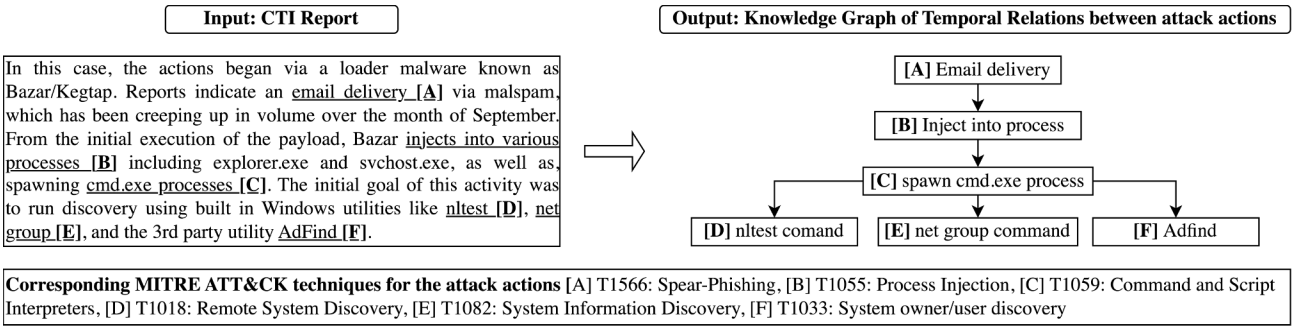

Fig. 1: An example of knowledge graph of temporal relations from a CTI report [1] describing Ryuk Ransomware attack

American English annotated with temporal information, and (b) TempEval-3 [3], a corpus annotated with temporal relations between events. TempEval-3 annotation uses TimeML (Section III-E1) to specify temporal relations. Researchers proposed several machine learning and natural language-based models to classify the temporal relations on the TempEval-3 dataset: (a) TARSQI [4], (b) CAEVO [5], (c) CATENA [6], and (d) Clear-TK [7]. The F1 score of these tools in TimeBank, TempEval-3 lies between 0.4 0.6 and, indicating that temporal extraction is a difficult task [8]. Temporal relation extraction between events has been applied on the domain-specific corpora, such as clinical procedures related text [9]. However, no such study exists for the cyberattack domain.

Cyberattack information extraction Researchers have contributed to natural language processing and machine learning techniques to identify the following cybersecurity-related information from CTI reports. Primarily, three types of information are extracted. First, extracting the indicators (such as malware hash, botnet IP address, file URL, vulnerability information) from text, which is relatively earliest among the three [10], [11]. Second, extraction of tactics, techniques, and procedures from CTI reports [12]–[17] have been investigated for the past several years, either as a text classification task or information extraction task. Third, of late, cybersecurity-related knowledge graph generation from CTI reports has gained attention from researchers [18]–[22]. Researchers have also performed detailed systematic literature reviews on information extraction from CTI reports [23]–[25]. The co-occurrence of techniques observed in past cyberattacks has also been investigated in [26]–[28]. Researchers have utilized the combinations and co-occurrences of attack actions to correlate intrusion alert logs, system logs, and kernel API calls to minimize false positives [29]–[34].

Overall, the aforementioned work extracts diverse types of information from text. However, none have extracted the cyberattack-related information from the time aspect. Identifying temporal information can provide an additional context to understanding how attack actions are performed one after another, what are happening together, and what actions are most likely to happen next. Hence, in this work, we contribute further to the domain of actionable CTI extraction by mining knowledge graph of temporal relations from CTI reports.

III. ChronoCTI Pipeline

This section discusses the process for constructing the ChronoCTI pipeline, as shown in Fig. 2.

A. Select a vocabulary of attack actions

We first choose a vocabulary for known attack actions. Several such vocabularies exist: (a) MITRE ATT&CK, (b) Cyber-kill-chain [35], (c) VERIS [36], and (d) CAPEC [37]. We choose MITRE ATT&CK as the vocabulary for the following reasons. MITRE ATT&CK is well known to cybersecurity practitioners, and MITRE actively collaborates with practitioners to update its vocabulary and repository [38]. MITRE ATT&CK is also used to specify attack actions in CTI reports. The cyber-kill chain focuses only on high-level actions, which is too abstract for our task. VERIS is a relatively newer framework, and CAPEC is more tailored to application security weaknesses. We use version 12.1[2] of MITRE ATT&CK containing 193 techniques.

To represent adversary action, ATT&CK uses two terms called *techniques*, and *procedures* [39], [40]. *Techniques* refer to an abstract action through which adversaries fulfill their goal [41], [42]. For example, the adversary can obtain their goal of accessing credentials by capturing user input, known as *T1056: Input Capture* [43] in ATT&CK. *Procedures* refer to the textual description of how the technique is implemented from actual cyberattack incidents [42]. For example, "FlawedAmmyy can collect keyboard events" [44], described as a procedure for the *T1056 input capture* technique. ATT&CK contains an enumeration of techniques and an enumeration of related procedure(s). All techniques' definitions and procedures are available.[3]. For each case of a group or malware using technique(s), ATT&CK contains a link to the CTI reports where an instance of a technique usage is documented, which we refer to as a *citation*. Fig. 1 shows the corresponding ATT&CK techniques used in the attack described in the excerpt from a CTI report.

B. Construct a dataset of attack actions in text

To predict attack action from text, we formulate the prediction as a text classification task, where a machine learning

[2]https://github.com/mitre/cti/releases/tag/ATT%26CK-v12.1
[3]https://attack.mitre.org/techniques/

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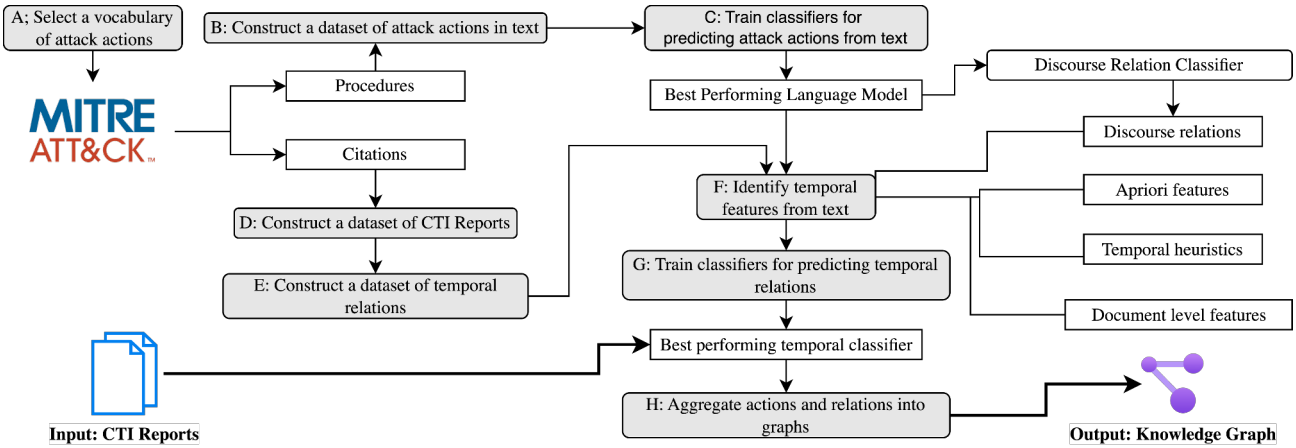

Fig. 2: Overview of ChronoCTI pipeline

model predicts the corresponding ATT&CK techniques given a piece of text. For example, consider the following sentence, *that those documents were likely delivered through spear-phishing attachments in emails* [45]. We need to build a classifier to predict the described ATT&CK technique in the sentence. The technique is *T1566: Phishing* in this specific case. We obtain the procedure enumeration from the definitions of ATT&CK techniques to train the text classifier. We use the procedure examples as training sentences and ATT&CK techniques as their corresponding labels.

However, we observed the following challenges in this prediction task. First, a sentence may not describe any technique at all. For example, *All of these factors align to point to the [...] Oilrig group as the likely threat actors behind this intrusion* [46]. Second, a sentence can describe more than one technique. For example, *A scheduled task was then created to assist in the execution of the keylogger* [46]. Here, we identify two techniques: *T1053: Scheduled Task/Job*, and *T1056: Input Capture*. However, procedure examples in ATT&CK are a one-to-one mapping between a sentence and a technique. Third, to train a text classifier, we must provide sufficient examples. However, ATT&CK lists 193 techniques, but not all have sufficient examples. We formulate the prediction task as a multi-class, multi-label text classification task to address the first two challenges. We only include techniques with at least 20 examples to address the third challenge. Although, as discussed in Section II, several works related to ATT&CK technique classification from text exist, we find only one work [16], which performs multi-label classification, but at the paragraph level, not the sentence level. Hence, we choose to construct the multi-label dataset from scratch and train learners on the dataset. We obtained 11,106 one-to-one mappings between sentences and 120 techniques from MITRE ATT&CK, having at least 20 examples.

We then conduct a round of mapping sentences to techniques to address the issue of missing multi-label mappings obtained from ATT&CK. We first apply c-TFIDF [47] as a weak classifier to get a prediction of one or more techniques given a sentence. We included this step because, during labeling, a person is required to process a large amount of information related to 120 techniques, which creates the risk of cognitive load, and c-TFIDF performs as a mitigating intervention. The first author then reviews (a) the sentence, (b) the prediction made by c-TFIDF, and (c) adds any missing mapping and rectifies any incorrect prediction from c-TFIDF. Then, the fourth author does the same on a randomly sampled 15% of the procedure examples. Then, the authors resolve their disagreements. We also include the original mapping as provided in ATT&CK. Thus, we obtain a multi-class, multi-label dataset. The two authors obtained a median Cohen's kappa (κ) score of 0.83 on the random sample, and according to [48], the score indicates a substantial agreement.

C. Train classifiers for predicting attack actions from text

We experimented with the following language models for the classifier on the multi-label dataset: (a) Roberta, a state-of-the-art transformer-based large language model from BERT trained on large web corpora by Google. We obtain the model from the Huggingface[4] repository. (b) CTI-Roberta: We fine-tuned a Roberta model on 3,263 articles cited in MITRE ATT&CK. We use *simpletransformer*[5] package and fine-tuned the model with default hyper-parameters, with batch sizes of 8, 16, and 32. We then select the model with the lowest cross-entropy loss from the three models trained from three batch sizes. (c) OpenAI: We fine-tuned the GPT3.5 text generation model (*gpt-3.5-turbo-1106*)[6] for prediction. The model is hosted on their own systems and uses default hyperparameters. (d) ChatGPT: We used *ChatGPT*[7] prompts to predict ATT&CK techniques given a sentence. We then compare the performance of the language models. We use macro F1 for comparison, which is the average F1 score for all labels [49]. We use the best model for the next steps.

[4]https://huggingface.co/roberta-base
[5]https://github.com/ThilinaRajapakse/simpletransformers
[6]https://platform.openai.com/docs/guides/fine-tuning
[7]https://openai.com/chatgpt/

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D. Construct a dataset of CTI reports

1) Select a source of CTI reports: To mine temporal relations from CTI reports, we need a dataset of reports. We use the website of MITRE ATT&CK as the single source for collecting CTI reports. We do so because the ATT&CK is maintained by collaborating with practitioners from multiple security vendors. Hence, any citation used in MITRE ATT&CK indicates that the corresponding CTI reports are of acceptable quality and sufficient technical description.

2) Filtering criteria: Our purpose is to mine temporal relations from reports. Hence, we only need reports that describe an attack incident written as a story with a sequentially ordered event narrative. Hence, we establish inclusion and exclusion criteria for filtering the citations so that we only have citations for CTI reports that describe cyberattack incidents in temporal order. **Inclusion criteria**: Content of a citation URL: (a) must be in English; (b) must be accessible online; and (c) describes cyberattack incidents step-by-step as time progresses. **Exclusion criteria** Content of a citation URL: (a) is detected by the browser as insecure; (b) does not contain cyberattack description; and (c) is a social media post, definition of a cybersecurity concept (e.g., Wikipedia, NIST glossary), API documentations, source code repositories, (d) describes malware features or cybercriminals capabilities, (e) does not describe an attack in a time-progression order, (f) non-attack relevant content such as countermeasures, security tools, statistics, impact statement, generic security news, and (g) is a pdf, presentation slides or audio/video media documents. Although PDF documents may be a good source for describing attacks in temporal order, we discarded them as the PDF documents are often a periodic compilation of multiple attacks published as web articles in the past, which we would have already obtained from the inclusion criteria.

After applying the criteria, we have a set of 673 CTI reports from 3,263 total citations from MITRE ATT&CK that contain attack descriptions as a sequence of events as time progresses. We refer to the set of reports as *SET-I*.

3) Construct an evaluation set of CTI reports: We found a portion of the reports in *SET-I* mentions the used ATT&CK techniques. We refer to the portion as *Set A*. Among *Set A*, we found a set of reports from a specific security vendor named *The DFIR Reports*[8]. The vendor's reports included the observed techniques and provided a graphical timeline of events, which helped build the dataset. We included all the CTI reports if they provided a list of techniques at the end of the report and also fulfilled all the inclusion and exclusion criteria. We refer to newly added reports as *Set B*. We then combine *Set A* and *Set B*, which we refer to as the *(SET-E)*. We use the reports from the set for building the dataset because the list of ATT&CK techniques provided at the end helps us identify the techniques and keeps us from missing any techniques.

SET-E contains 54 reports from *SET-A* and 40 reports from *SET-B*, resulting in 94 reports consisting of 7,052 sentences.

[8]https://thedfirreport.com/

E. Construct a dataset of temporal relations

1) Defining Temporal Relation between Attack Action: A well-known taxonomy for specifying temporal relations among events in the text is required to represent the temporal relation between actions. Literature has such taxonomy named TimeML [50], which defines 14 temporal relations (known as *TLINK*) in their latest specification v1.2.1 [51], form where we use the following five: (a) before, (b) after, (c) immediately before, (d) immediately after, and (e) simultaneous. The relations (a) - (d) reflect two events happening one after another, which we refer to as the BEFORE relation in this work. However, relation (e) can reflect two different cases. **C1**, two actions happen at the same time, and the implementation of the two actions overlaps with each other. E.g., action B and C in Fig. 1. **C2**, two actions happen simultaneously but independently. For example, actions D, E, and F in Fig. 1. We refer to cases (C1) and (C2) as SIMULTANEOUS-OVERLAP (SIM-OV) and CONCURRENT, respectively. Finally, we use a temporal relation named NULL to represent no relation between two actions. We summarize the relations in Table I.

2) Ground truth construction: We define predicting the temporal relations of attack actions as the following. Let, a CTI report R may contain one or multiple techniques from the selected 120 ATT&CK techniques: $T_1, T_2, ..., T_{120}$. We then construct a set of all possible pairs between the 120 techniques. If, in R, we observe that the two techniques in the set of pairs are executed concerning time progression, then the pair has a temporal relation between them. Our task is to train a model that takes R as an input and predicts temporal relations among all possible pairs of techniques. The first three authors manually investigate all 94 reports from *SET-E* and to construct a ground truth dataset of temporal relations. The first author reviews all reports. Four reports were used as examples for the second and third authors. Among the rest, the second author reviews 69, and the third reviews 19 reports. After reviewing, the authors resolve their disagreements.

The dataset contains 1,342,320 rows, each representing a pair between techniques and the corresponding temporal label. CONCURRENT, BEFORE, and SIM-OV have been mapped to 179, 1,208, and 1,031 rows. The rest of the rows are mapped to NULL. The Cohen's κ agreement scores of CONCURRENT, BEFORE, and SIM-OV are 0.49, 0.47, and 0.47, respectively, which is moderate agreement [48].

F. Identify temporal features from text

We need to identify features to train a learner on the features to predict temporal relations among pairs of techniques. Assume we want to identify features of a pair of techniques (T_x, T_y) in a CTI report R. We describe the features below. The features are numeric attributes.

1) Default feature: Probability of T_x, and T_y in sentences of R: If there is a relation between T_x and T_y, these two techniques should be present in R, and the classifier from Section III-C would predict the techniques with higher probability. If R contains m sentences, we calculate the probabilities

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TABLE I: List of temporal relations

Relation	Definition, example, and explanation
BEFORE	**Definition:** two techniques happen one after another sequentially as time progresses. **Example:** *Upon opening the file, the **user was prompted to enable macros** to complete the form, which began the execution of the malware. Once executed, the macro **created a VBS script** (Updater.vbs), two PowerShell scripts (temp.ps1 and Script.ps1), and installed persistence through a scheduled task.* **Explanation:** Here, *T1059: Command and Script Interpreter* technique is performed after *T1204: User Execution*. Hence, $(T1204, T1059)$ have a *BEFORE* relation.
SIM-OV*	**Definition:** two techniques are executed together and one requires the other for execution **Example:** *A **scheduled task** was then created to assist in execution of the **keylogger**.* **Explanation:** Here, *T1053: Scheduled Task* technique is performed to facilitate the execution of *T1056: Input Capture*. Hence, $(T1053, T1056)$ have an *SIMULTANEOUS-OVERLAP* relation.
CONCURRENT	**Definition:** two techniques executed independently, however, happen at the same phase of an attack. **Example:** *all of which were executed via PowerShell cmdlets or built-in Windows utilities like **whoami**, net, **time**, tzutil and tracert;* **Explanation:** Here, *T1033: System User/Owner Discovery*, and *T1124: System Time Discovery* are executed independently, but they happen during the discovery phase of an attack. Hence, $(T1033, T1124)$ have an *CONCURRENT* relation.
NULL	**Definition:** No temporal relation exists between two techniques **Example:** *all of which were executed via PowerShell cmdlets or built-in Windows utilities like **whoami**, net, **time**, tzutil and tracert;* **Explanation:** This example does not discuss *T1566: Phishing*, and *T1204: User Execution*. Hence, in the context of this sentence, $(T1204, T1566)$ has a *NULL* relation.

SIM-OV: SIMULTANEOUS-OVERLAP

TABLE II: List of temporal markers

Relation	Temporal marker
BEFORE	after, afterward, following, immediately, instantly, later, next, then, succeeding, subsequent, subsequently, before, previous, prior, previously, preceding
SIM-OV	during, while, within, through, throughout
CONCURRENT	concurrent, concurrently, contemporary, simultaneous, simultaneously

of T_x and T_y for m sentences, and then we take the top five probabilities for T_x and T_y as features.

2) F1: Temporal heuristics: (a) **Temporal markers in text:** Natural language often contains specific words that can denote the presence of temporal relations, such as *before*, *after*. We obtain a set of temporal markers from [52] shown in Table II and use them as heuristics. (b) **TimeML features:** *TimeML* is a specification to specify temporal event sequences in text. For example, the sentence *Upon opening the file, the user was prompted to enable macros* [46] can be specified using *TimeML* as following: *event1=open, event2=enable, relation=before*. We use *TARSQI*[9] to get the TimeML annotations of events from a report. For each event identified by TARSQI, we compute the surrounding noun or verb phrases and then find the corresponding techniques. We then record the event relation types, as defined in TimeML. We then aggregate the TimeML relation types per pair of techniques and record counts of relations seen for each pair of techniques.

3) F2: Document level feature: (a) **Sentence adjacency:** If T_x and T_y are related temporarily, the corresponding sentences should be closely located. We count the occurrences where two sentences, with a distance $d = [-4, 4]$, contain T_x and T_y, respectively. $d = 0$ for two sentences next to each other. $d = 1$ means another sentence separates two sentences. (b) **Same sentence:** If T_x and T_y are related temporarily, they may share the same sentence. We compute the count of sentences where T_x and T_y are found. (c) **Sentence similarity:** If T_x and T_y are temporarily related, the sentences containing the two techniques may have a higher semantic similarity. Using word embeddings, we compute the cosine similarity [53] between sentences where T_x is present and where T_y is present. For the embedding, we use Gensim to train a custom word embedding based on 3,263 articles from ATT&CK. (d) **Coreference:** If T_x and T_y are related temporarily, then the sentences containing the two techniques may have a coreference relation [54]. E.g., *Upon opening the file, the user was prompted to enable macros to complete the form, which began executing the malware. Once executed, the macro created a VBS script* [46]. Here, *the macro* is coreferencing the *macros* stated in the first sentence. We use *neuralcoref*[10] deep learning model to identify the sentence's coreference relations.

4) F3: Discourse relations: Discourse relations indicate the rhetorical structure among the sentences in a paragraph or article [54]. Thus, discourse features can indicate temporal relations between techniques indicated among sentences. We use five discourse relations from [54] to indicate if there is any temporal relation between the first and second sentences: S1 and S2, respectively. (a) **Next:** S2 describes something that happened after the events described in S1. For example, *Upon opening the file, the user was prompted to enable macros to complete the form, which began the execution of the malware. Once executed, the macro created a VBS script (Updater.vbs), two PowerShell scripts (temp.ps1 and Script.ps1), and installed persistence through a scheduled task* [46]. Here, a user first executes a macro (T1204: User Execution) in S1, and a script is then executed (T1059: Command and Script) in S2. (b) **Elaboration:** S2 explains the action stated in S1. E.g., *The intrusion began with the execution of a malicious macro within a Word document. The document was themed as a job application for the firm Lumen* [46]. Here, a user first execute a macro (T1204: User Execution) in S1, however, S2 does not indicate what happens next, rather further explain from where the macro came. (c) **If-else:** S2 describes a conditional

[9]http://timeml.org/site/tarsqi/toolkit/

[10]https://github.com/huggingface/neuralcoref

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outcome of an action described in S1. E.g., *if the current user has high privileges, install.bat is executed directly; Otherwise, mshlpweb.dll is executed using rundll32.exe* [55]. Here, the execution of rundll32.exe (T1218: System Binary Proxy Execution) in S2 depends on the execution of install.bat (T1059: Command and Script). (d) **List:** S1 and S2 have similar semantic meanings - however, contain independent neucleus[11]. E.g., *Install.bat — acts as an installer to ensure persistence and execute mshlpsrvc.dll, mshlpweb.dll — acts as loader; responsible to elevate privileges* [55]. Here, S1 and S2 do not indicate any temporal relation. (e) **Miscellaneous:** S1 and S2 do not have any discourse relation, or they have a discourse relation other than the previous four. E.g., *Constructs and executes the command-line to download additional files, The document contains three hidden text boxes* [55].

We train a discourse relation classifier, which we model as a sentence-pair classification task. We create a dataset of sentence pairs and their corresponding discourse relations. The pairs came from (a) the sentences located one after another in CTI reports and (b) the sentences having coreferences [54]. The first and the third author independently annotate the dataset. Then, the authors resolve their disagreements. On the constructed dataset, we train the classifier using a deep neural network with a transformer architecture based on the CTI-Roberta language model proposed in Section III-C. We use the *simpletransformer* python package for the task.

5) F4: Apriori features: We assume, if T_x and T_y are related temporarily, they were seen together in past cyberattacks. We can obtain such information from MITRE ATT&CK's [42] repository of malware[12], cybercrime groups[13], and past campaigns[14]. We obtain the information and apriori features by performing association rule mining [56]. We obtain numeric values of the following well-known association rule mining scores: (a) support, (b) pointwise mutual information, (c) phi correlation coefficient, (d) causal support, (e) Jaccard distance, (f) confidence, (g) causal confidence, (h) conviction, and (i) added value [57]. We also performed one-hot categorical embedding [58] by representing the numeric values of apriori features associated with each pair of techniques.

TABLE III: Comparison of language models

Language model	Precision	Recall	F1
Roberta Base	0.82	0.72	0.75
Roberta CTI	0.82	0.73	0.76
OpenAI GPT-3.5-1106	0.64	0.62	0.61
ChatGPT	0.33	0.33	0.29

G. Train classifiers for predicting temporal relations

We train three types of supervised classifiers on temporal features (default and F1-F4). (a) **Traditional classifiers** primarily refers to the non-deep neural network-based classifiers: Gradient boosting (GB), Random forest (RF), Gaussian Naive Bayesian (GNB), K nearest neighbors (KNN), Stochastic Gradient Descent (SGD), and Decision tree (DT) classifiers. We use the classifiers from the *scikit-learn* packages[15]. (b) **Deep Neural Network based classifiers** Traditionally, deep classifiers do not perform well for a small number of features [59], reflecting our case. Hence, we use a deep neural network classifier proposed by Google named *TabNet* [59], which is suited for a low number of features and includes both the functionalities of transformers and the attention mechanism. We use *pytorch* package[16] for the task. We also use the multi--layer perception (MLP) classifier from *scikit-learn*. (c) **Graph Neural Net Classifiers** The temporal relation prediction task can also be formulated as a graph where techniques are nodes, pairs of techniques are edges, and all the identified features are numeric edge attributes. Then, the task can be formulated as an edge label prediction problem. We use the relational graph convolutional neural network (RGCN) classifier as a predictor [60], and we use *Deep graph library*[17] for performing the task. We evaluate the classifiers on the following metrics[18]: precision (P), recall (F), and F1 score (F). However, as the classifiers predict one or more temporal relations for a pair of techniques, we use the macro average of P, R, and F, which are the averages of P, R, and F of the four temporal relation types. We also use *precision@k (P@K)*, and *normalized discounted cumulative gains (NDCG)* metrics especially designed for multi-label classification[19], which denote how well the model can rank the relevant patterns before the irrelevant items based on their prediction probability.

H. Aggregate actions and relations into graphs

From Step G, for each report, we obtain a set of techniques and a set of temporal relations among the techniques. We construct a graph for each report where techniques are nodes and relations are edges. To evaluate the knowledge graph construction performance, we use the MUC-5 [61] augmented precision, recall, and F1 score, a common metric to evaluate the performance of relation extraction tasks between nodes, such as co-reference resolution [62], and named entity recognition [63]. Since our task is to extract relations between two techniques, similar to the relation extraction between two nodes (assuming techniques as nodes), MUC-5 is a good evaluation metric for the case.

IV. Evaluation of ChronoCTI

This section describes the evaluations of ChronoCTI.

A. Predicting attack actions from text

We train the CTI-Roberta model from the Roberta base model. We identified the lowest loss in training the Roberta

[11] neucleus refers to the protagonist or antagonist in a sentence
[12] https://attack.mitre.org/software/
[13] https://attack.mitre.org/groups/
[14] https://attack.mitre.org/campaigns/
[15] https://scikit-learn.org/stable/modules/classes.html
[16] https://pytorch.org/
[17] https://www.dgl.ai/
[18] https://scikit-learn.org/stable/modules/classes.html#classification-metrics
[19] https://scikit-learn.org/stable/modules/classes.html#multilabel-ranking-metrics

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TABLE IV: Classification performance, and inter-rater agreement for the discourse relation classifier, and the discourse relation dataset respectively

Discourse relation	Pair type	Count	P	R	F	κ score
ELABORATION	Adjacent	2087	0.77	0.87	0.82	0.49*
	Coreferenced	762	0.58	0.65	0.61	
IF_ELSE	Adjacent	81	0.89	0.89	0.89	0.61**
	Coreferenced	21	0.75	0.60	0.67	
LIST	Adjacent	245	0.87	0.63	0.73	0.56*
	Coreferenced	41	1.00	0.20	0.33	
MISC	Adjacent	1258	0.68	0.74	0.71	0.47*
	Coreferenced	1163	0.72	0.71	0.72	
BEFORE	Adjacent	1278	0.83	0.62	0.71	0.40*
	Coreferenced	672	0.67	0.63	0.65	

*moderate agreement, **substantial agreement [48]

CTI with a batch size of 16. For Roberta Base and Roberta CTI, we use Spacy's default deep neural network architecture[20] except for maximum steps and patience - which are set to 5,000 and 750, respectively. To fine-tune the OpenAI GPT-3.5, we use the default hyper-parameters. In Table III, we compare the five models after performing five-fold cross-validation with a split of 80% training and 20% testing set. CTI-Roberta model performs best and slightly outperforms the Roberta base model in classifying a sentence to multiple techniques. GPT-3.5 fine-tuned model ranks third, scoring 20% less than Roberta models. However, the worst-performing model in our experiment is using ChatGPT from OpenAI, which uses zero-shot learning. The other three models achieved more than twice the F1 score achieved by ChatGPT, with the Roberta-CTI model scoring the best. **Hence, we choose CTI-Roberta as the language model for the next steps.**

B. Temporal features from text

Section III-F discussed five types of feature sets, which results in 309 numeric features. The number of features per feature set are: (a) default feature = 10, (b) F1 = 17, (c) F2 = 13, (d) F3 = 10, and (e) F4 = 259. Section III-F4 discussed the methodology for identifying a set of five discourse relations between sentences. The discourse relation dataset contains 7,598 sentence pairs of 4,944 adjacent and 2,694 coreferenced sentence pairs. These sentence pairs are collected from train split from *SET-E*, using 73 reports for training and 21 reports for evaluation. We trained two classifiers for discourse identification, one for the adjacent sentence pairs and another for the coreferenced sentence pairs. We show the classifier performance and inter-rater agreement score in Table IV.

C. Temporal relation classifier

We train the selected classifiers (Section III-G) on the *SET-E* consisting of 94 reports. Among the 94, 73 reports were used as the training set, and 21 reports were used as the evaluation set. The training set contains 10423544 NULL, 908 NEXT, 711 SIM-OV, and 119 CONCURRENT relations. The test set contains 301720 NULL, 300 NEXT, 320 SIM-OV, and 60 CONCURRENT relations.

[20]https://spacy.io/usage/processing-pipelines

TABLE V: Performance of Temporal Relation Classifiers

Classifier	Features*	P	R	F
Gradient Boosting (GB)	F1+F2+F3+F4	0.87	0.61	**0.71**
Decision Tree (DT)	F1+F2+F3+F4	0.64	0.64	0.64
Random Forest (RF)	F1+F2+F3+F4	**0.9**	0.54	0.64
MLP	F1+F2+F3+F4	0.86	0.51	0.61
Stochastic Gradient Descent (SGD)	F1+F2+F3+F4	0.87	0.52	0.61
RGCN	F1+F2+F3+F4	0.74	0.51	0.58
K-nearest Neighbors (KN)	F1+F2+F3+F4	0.76	0.48	0.56
RGCN	F1+F2	0.64	0.41	0.48
RGCN	F1	0.67	0.4	0.45
RGCN	F1+F2+F3	0.63	0.39	0.44
Decision Tree	F1+F2+F3	0.44	0.43	0.43
Decision Tree	F1+F2	0.4	0.42	0.41
Gradient Boosting	F1+F2	0.53	0.38	0.4
Gradient Boosting	F1+F2+F3	0.5	0.38	0.39
Gradient Boosting	F1	0.43	0.37	0.38
K-nearest Neighbors	F1+F2	0.49	0.37	0.38
K-nearest Neighbors	F1	0.45	0.37	0.38
Random Forest	F1+F2	0.57	0.37	0.38
Random Forest	F1+F2+F3	0.57	0.37	0.38
TabNet	F1+F2+F3+F4	0.55	0.37	0.38
MLP	F1+F2+F3	0.39	0.36	0.37
TabNet	F1	0.38	0.36	0.37
Stochastic Gradient Descent	F1+F2+F3	0.38	0.37	0.37
Stochastic Gradient Descent	F1+F2	0.38	0.37	0.37
Stochastic Gradient Descent	F1	0.38	0.37	0.37
TabNet	F1+F2	0.38	0.37	0.37
Random Forest	F1	0.46	0.37	0.37
K-nearest Neighbors	F1+F2+F3	0.49	0.37	0.37
MLP	F1	0.37	0.36	0.37
MLP	F1+F2	0.38	0.37	0.37
TabNet	F1+F2+F3	0.37	0.35	0.36
Decision Tree	F1	0.35	0.38	0.36
GaussianNB (GNB)	F1+F2	0.31	0.87	0.35
GaussianNB	F1	0.31	0.84	0.34
GaussianNB	F1+F2+F3	0.28	**0.89**	0.31
GaussianNB	F1+F2+F3+F4	0.28	0.81	0.25

*Default is included in all combinations of F1-F4

On the 73 reports, we first performed a five-fold cross-validation of all the selected classifiers across all the feature sets (Section III-F). Note that, for all feature combinations, we used the default feature set with additional combinations of features. Table V shows the median of the classifiers' macro precision, macro recall, and macro F1 scores across all four feature sets. We observe that the GB classifier performs best. We also observe GB and RF perform better than MLP, TabNet, RGCN, KNN, and SGD. Specifically, GB perform at least 10% better than the best-performing deep neural network model. Although deep neural network-based models perform better with complex and large feature sets, literature shows [59] that gradient-boosting learners perform the best with the tabular feature type. In our experiment, we observe the same. Moreover, ensemble learners utilize the overall predictions from multiple base learners, which may work well on noisy features. On the other hand, MLP, TabNet, RGCN, KNN, and SGD learners use a single base model, which may have been impacted more by the noisy features. In our case, features related to technique prediction, TimeML, and discourse relations are noisy because their value is derived from classifiers in previous steps before temporal classification. Hence, ensemble learners GB, and RF performed better

The apriori features (F4) significantly contribute to the

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TABLE VI: Performance of Extreme Gradient Boosting classifier on the evaluation dataset

Labels	P	R	F	P@50	P@100	S	AUC
CONCURRENT	0.77	0.33	0.47	0.77	0.77	60	0.84
BEFORE	0.65	0.34	0.45	0.76	0.72	300	0.91
NULL	0.97	0.99	0.98	1.00	1.00	301720	0.91
SIMU-OV.	0.61	0.17	0.27	0.72	0.61	320	0.91
Macro Avg.	0.75	0.46	0.54	0.81	0.78	302400	-
NDCG				0.98			

TABLE VII: Performance after Handling Class Imbalance

Classifier + Technique	Type	P	R	F
GB + KMeansSMOTE	OS	0.71	0.51	0.57
GB + SVMSMOTE	OS	0.64	0.50	0.55
GB + Unbalanced	-	0.75	0.46	0.54
GB + OneSidedSelection	US	0.74	0.46	0.54
GB + SMOTETomek	OS+US	0.63	0.50	0.54
GB + SMOTE	OS	0.60	0.49	0.53
GB + SMOTEENN	OS+US	0.56	0.54	0.53
GB + BorderlineSMOTE	OS	0.64	0.47	0.52

OS: oversampling, US: undersampling

classifiers. Without the features, learners perform well in the recall score. However, the feature helps improve the precision score in all the classifiers. Without the apriori feature, GNB performs the best. We observe the best precision score from the RF. The best recall score is observed from the GNB. However, GB performed the best neither in precision nor recall. However, the classifiers score best in the case of the F1 score. Overall, with all the features, the top-performing models perform very well in precision but lack recall.

We applied the best-performing model: the gradient boosting classifier on the evaluation dataset of 21 reports from *SET-E*. We summarize the performance in Table VI. We observe the following from the table. First, the classifiers perform almost perfectly for the NULL class, but since, NULL class constitutes 99% of the dataset, such performance is expected. However, the classifier shows decent precision for the BEFORE, CONCURRENT, and SIM-OV classes but low recall. These three classes constitutes less than 1% of the whole dataset, and hence, suffers the low recall due to class imbalance issues. The classifier predicts with accuracy, as confirmed by higher precision, however, the model misses a lot of true examples, as shown by low recall scores. Second, the precision@50 and precision@100 metrics for all the classes show a decent score, which suggests that the top 50 or top 100 predictions made by the classifier are often accurate. Finally, the NDCG scores are very close to 1.00, which suggests that if we rank the predictions by their probability, top predictions will be correct about 98% of the time. However, from the support column, we also observe that the 99% majority of the NULL class skews the macro F1 and NDCG.

Table VI shows that the dataset is severely imbalanced. Only 0.22% of the dataset contains a non-NULL relation. We experimented with the oversampling and undersampling and combined sampling techniques[21], shown in Table VII, to improve the classification performances. We observe that only one oversampling technique, K-Means SMOTE, improves the performance, although by only 6%.

The evaluation split of the dataset (21 reports) contains 169 instances of temporal relations that were not present in the train-test split (73 reports). Among the 169, 112 are from BEFORE, 42 from SIM-OV, and 15 are from CONCURRENT pairs. We identify that the best-performing model identifies 10 from CONCURRENT and 6 from BEFORE pairs. The model identifies 16 (9.5%) of the unseen pairs. The observation suggests that the learner handles unseen CONCURRENT relations well and can identify a few of the BEFORE relations but cannot identify any unseen SIM-OV relations. Despite the low performance, the model performs decently given the context of extreme class imbalance.

From Table VI and the observation with unseen relations, **ChronoCTI** performs best for detecting CONCURRENT and BEFORE relations. However, it performs poorly in the detection of SIM-OV relations. Class imbalance issue inherent in the dataset impacts the performance of the model.

D. Evaluate graph level performance

After performing Step H on the evaluation dataset, we obtain 21 knowledge graphs[22]. We compute the MUC-5 precision, recall, and F1 score: 0.58, 0.25, and 0.34. Like relation classification performance, the knowledge graph construction task also shows better precision but a weak recall score.

V. Discussion and Conclusion

We propose ChronoCTI, which is the first automated pipeline that can input an unstructured text related to cyberattacks and output a knowledge graph of temporal relations among attack actions in cyberattacks. The nodes are MITRE ATT&CK techniques, a well-known vocabulary of attack actions. The edges are of three types: BEFORE, SIMULTANEOUS-OVERLAP, and CONCURRENT. ChronoCTI works in three primary steps. First, ATT&CK techniques are identified from text with a macro precision, recall, and F1 score of 0.82, 0.83, and 0.76. In the next step, we compute features for representing temporal relations and train classifiers where we observe a macro precision, recall, and F1 score of 0.75, 0.46, and 0.54. In the third step, we observe the MUC-5 precision, recall, and F1 score on the constructed knowledge graphs: 0.58, 0.25, and 0.34. Overall, ChronoCTI demonstrates good precision but low recall. We discuss several factors that impact the performance of ChronoCTI, followed by threats to validity and future research directions.

Further discussion of the overall performance First, we use a classifier (ATT&CK technique) in step C and two classifiers in step F (TARSQI and discourse). All three classifiers have an F1 score < 0.80. The temporal classifier in step G depends on the outputs of these three classifiers and, is

[21] https://imbalanced-learn.org/stable/references/index.html

[22] generated graphs can be accessed online, links are available in the replication package, e.g., see https://poloclub.github.io/argo-graph-lite/#6fbd35a6-3bb4-4b9a-849c-02d470c172af for the report [46]

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impacted by the false positive and negative outputs. Literature shows that these three classifiers are still an active research topic, and even the best models still lack a good F1. Second, we observe the issue of class imbalance while training the temporal learner. The NULL relation is the majority class, with 99% of the relation types introducing a severe imbalance in the dataset. However, the imbalance issue can not be resolved easily because a report only contains a few possible relations among all possible relations among 120 techniques. Moreover, the class imbalance problem also affected the classifier of step C. Among the selected 120 techniques, 2 have more than 1000 examples, 4 have examples between 500-1000, 49 have 100 to 500, and the rest 65 have less than 100 examples. Third, ATT&CK techniques can be easily understood by humans but not by models due to the use of technical words not in the training dataset. Fourth, the proposed pipeline consists of several components, such as technique classifier, temporal parsing, relation classifier. Each component required manual effort for annotation, and run on its own dataset. Hence, the combined model requires a decent amount of computation resource and time. Finally, the performance could have been better if the ground truth dataset for temporal relations was bigger. Nonetheless, the authors needed about an hour to identify all temporal relations per report.

Threats to validity In Steps B, D, and E, we built ground truth subject to first author's judgment. We account for this by including the second, third, and fourth authors in constructing the dataset. We acknowledge the dataset may depend on the authors' domain expertise. We used inclusion and exclusion criteria to select CTI reports with acceptable verbosity. However, the filtering led to fewer reports. In literature, several studies [17], [19] in extracting CTI and temporal relations have used similar sample sizes. Our work does not investigate all temporal relations specified in TimeML, and all ATT&CK techniques. However, we provide a methodology to replicate and engineer the features for temporal relations and existing techniques to be introduced in future versions of ATT&CK or other taxonomies. ChronoCTI shows lower recall, indicating that ChronoCTI missed many true positives in Step G. However, the higher precision of ChronoCTI results in fewer false alerts. Finally, with the change in threat landscape, the model may not perform well against unseen relations. The issue is pre-dominant in all kinds of ML models, and can be addressed with the inclusion of new dataset on latest CTI reports.

Future research direction First, ChronoCTI has higher precision and lower recall, which motivates further investigation into improving the recall through a bigger dataset, addressing class imbalance, and incorporating techniques from future releases of ATT&CK. To improve the performance, every components of ChronoCTI should be trained in higher volume of datasets. Second, we suggest introducing threat models into the application development process so that they can specify how actions follow one another. This would aid practitioners in clearly and concisely representing and sharing the chain of attack actions. We advocate security application developers devise a threat-hunting mechanism based on the relations so that practitioners can identify potentially malicious behavior from system logs and execution traces. The generated knowledge graph could have been linked with different types of knowledge graphs available in the literature. Third, the graphs can be combined with one another two form a heterogeneous information graph. The combined graphs would facilitate querying, knowledge based reasoning, and constructing semantic models for threat hunting. Finally, alert fatigue is a commonly known issue in intrusion detection [33]. The identified patterns, paired with machine learning, can be investigated to reduce the number of false alerts.

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References

[1] "Ryuk's Return," https://thedfirreport.com/2020/10/08/ryuks-return/, 2014.

[2] J. Pustejovsky, P. Hanks, R. Sauri, A. See, R. Gaizauskas, A. Setzer, D. Radev, B. Sundheim, D. Day, L. Ferro *et al.*, "The timebank corpus," in *Corpus linguistics*, vol. 2003. Lancaster, UK, 2003, p. 40.

[3] N. UzZaman, H. Llorens, L. Derczynski, J. Allen, M. Verhagen, and J. Pustejovsky, "Semeval-2013 task 1: Tempeval-3: Evaluating time expressions, events, and temporal relations," in *Second joint conference on lexical and computational semantics (* SEM), volume 2: Proceedings of the seventh international workshop on semantic evaluation (SemEval 2013)*, 2013, pp. 1–9.

[4] M. Verhagen and J. Pustejovsky, "Temporal processing with the tarsqi toolkit," in *COLING 2008: Companion Volume: Demonstrations*, 2008, pp. 189–192.

[5] N. Chambers, T. Cassidy, B. McDowell, and S. Bethard, "Dense event ordering with a multi-pass architecture," *Transactions of the Association for Computational Linguistics*, vol. 2, pp. 273–284, 2014.

[6] P. Mirza and S. Tonelli, "Catena: Causal and temporal relation extraction from natural language texts," in *The 26th international conference on computational linguistics*. ACL, 2016, pp. 64–75.

[7] S. Bethard, "ClearTK-TimeML: A minimalist approach to TempEval 2013," in *Second Joint Conference on Lexical and Computational Semantics (*SEM), Volume 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013)*, S. Manandhar and D. Yuret, Eds. Atlanta, Georgia, USA: Association for Computational Linguistics, Jun. 2013, pp. 10–14. [Online]. Available: https://aclanthology.org/S13-2002

[8] M. Ocal, A. Perez, A. Radas, and M. Finlayson, "Holistic evaluation of automatic timeml annotators," in *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, 2022, pp. 1444–1453.

[9] Y. B. Gumiel, L. E. Silva e Oliveira, V. Claveau, N. Grabar, E. C. Paraiso, C. Moro, and D. R. Carvalho, "Temporal relation extraction in clinical texts: a systematic review," *ACM Computing Surveys (CSUR)*, vol. 54, no. 7, pp. 1–36, 2021.

[10] Z. Zhu and T. Dumitras, "Chainsmith: Automatically learning the semantics of malicious campaigns by mining threat intelligence reports," in *2018 IEEE European symposium on security and privacy (EuroS&P)*. IEEE, 2018, pp. 458–472.

[11] J. Liu, J. Yan, J. Jiang, Y. He, X. Wang, Z. Jiang, P. Yang, and N. Li, "Tricti: an actionable cyber threat intelligence discovery system via trigger-enhanced neural network," *Cybersecurity*, vol. 5, no. 1, p. 8, 2022.

[12] W. Ge, J. Wang, T. Lin, B. Tang, and X. Li, "Explainable cyber threat behavior identification based on self-adversarial topic generation," *Computers & Security*, vol. 132, p. 103369, 2023.

[13] V. Orbinato, M. Barbaraci, R. Natella, and D. Cotroneo, "Automatic mapping of unstructured cyber threat intelligence: An experimental study:(practical experience report)," in *2022 IEEE 33rd International Symposium on Software Reliability Engineering (ISSRE)*. IEEE, 2022, pp. 181–192.

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[14] Y. You, J. Jiang, Z. Jiang, P. Yang, B. Liu, H. Feng, X. Wang, and N. Li, "Tim: threat context-enhanced ttp intelligence mining on unstructured threat data," *Cybersecurity*, vol. 5, no. 1, p. 3, 2022.

[15] Y. Wu, Q. Liu, X. Liao, S. Ji, P. Wang, X. Wang, C. Wu, and Z. Li, "Price tag: towards semi-automatically discovery tactics, techniques and procedures of e-commerce cyber threat intelligence," *IEEE Transactions on Dependable and Secure Computing*, 2021.

[16] V. Legoy, M. Caselli, C. Seifert, and A. Peter, "Automated retrieval of att&ck tactics and techniques for cyber threat reports," *arXiv preprint arXiv:2004.14322*, 2020.

[17] G. Husari, E. Al-Shaer, M. Ahmed, B. Chu, and X. Niu, "Ttpdrill: Automatic and accurate extraction of threat actions from unstructured text of cti sources," in *Proceedings of the 33rd annual computer security applications conference*, 2017, pp. 103–115.

[18] K. Ahmed, S. K. Khurshid, and S. Hina, "Cyberentrel: Joint extraction of cyber entities and relations using deep learning," *Computers & Security*, vol. 136, p. 103579, 2024.

[19] Z. Li, J. Zeng, Y. Chen, and Z. Liang, "Attackg: Constructing technique knowledge graph from cyber threat intelligence reports," in *European Symposium on Research in Computer Security*. Springer, 2022, pp. 589–609.

[20] C.-C. Huang, P.-Y. Huang, Y.-R. Kuo, G.-W. Wong, Y.-T. Huang, Y. S. Sun, and M. C. Chen, "Building cybersecurity ontology for understanding and reasoning adversary tactics and techniques," in *2022 IEEE International Conference on Big Data (Big Data)*. IEEE, 2022, pp. 4266–4274.

[21] Y. Ren, Y. Xiao, Y. Zhou, Z. Zhang, and Z. Tian, "Cskg4apt: A cybersecurity knowledge graph for advanced persistent threat organization attribution," *IEEE Transactions on Knowledge and Data Engineering*, 2022.

[22] K. Satvat, R. Gjomemo, and V. Venkatakrishnan, "Extractor: Extracting attack behavior from threat reports," in *2021 IEEE European Symposium on Security and Privacy (EuroS&P)*. IEEE, 2021, pp. 598–615.

[23] M. R. Rahman, R. M. Hezaveh, and L. Williams, "What are the attackers doing now? automating cyberthreat intelligence extraction from text on pace with the changing threat landscape: A survey," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–36, 2023.

[24] X. Zhao, R. Jiang, Y. Han, A. Li, and Z. Peng, "A survey on cybersecurity knowledge graph construction," *Computers & Security*, p. 103524, 2023.

[25] M. R. Rahman, R. Mahdavi-Hezaveh, and L. Williams, "A literature review on mining cyberthreat intelligence from unstructured texts," in *2020 International Conference on Data Mining Workshops (ICDMW)*. IEEE, 2020, pp. 516–525.

[26] C. Shin, I. Lee, and C. Choi, "Exploiting ttp co-occurrence via glove-based embedding with mitre att&ck framework," *IEEE Access*, 2023.

[27] R. Al-Shaer, J. M. Spring, and E. Christou, "Learning the associations of mitre att & ck adversarial techniques," in *2020 IEEE Conference on Communications and Network Security (CNS)*. IEEE, 2020, pp. 1–9.

[28] M. R. Rahman and L. Williams, "Investigating co-occurrences of mitre att\&ck techniques," *arXiv preprint arXiv:2211.06495*, 2022.

[29] T. Chen, H. Zeng, M. Lv, and T. Zhu, "Ctimd: Cyber threat intelligence enhanced malware detection using api call sequences with parameters," *Computers & Security*, vol. 136, p. 103518, 2024.

[30] P. Gao, F. Shao, X. Liu, X. Xiao, Z. Qin, F. Xu, P. Mittal, S. R. Kulkarni, and D. Song, "Enabling efficient cyber threat hunting with cyber threat intelligence," in *2021 IEEE 37th International Conference on Data Engineering (ICDE)*. IEEE, 2021, pp. 193–204.

[31] A. Berady, M. Jaume, V. V. T. Tong, and G. Guette, "From ttp to ioc: Advanced persistent graphs for threat hunting," *IEEE Transactions on Network and Service Management*, vol. 18, no. 2, pp. 1321–1333, 2021.

[32] Y.-T. Huang, C. Y. Lin, Y.-R. Guo, K.-C. Lo, Y. S. Sun, and M. C. Chen, "Open source intelligence for malicious behavior discovery and interpretation," *IEEE Transactions on Dependable and Secure Computing*, vol. 19, no. 2, pp. 776–789, 2021.

[33] S. M. Milajerdi, R. Gjomemo, B. Eshete, R. Sekar, and V. Venkatakrishnan, "Holmes: real-time apt detection through correlation of suspicious information flows," in *2019 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2019, pp. 1137–1152.

[34] S. M. Milajerdi, B. Eshete, R. Gjomemo, and V. Venkatakrishnan, "Poirot: Aligning attack behavior with kernel audit records for cyber threat hunting," in *Proceedings of the 2019 ACM SIGSAC conference on computer and communications security*, 2019, pp. 1795–1812.

[35] L. Martin, "Cyber Kill-Chain," https://www.lockheedmartin.com/en-us/capabilities/cyber/cyber-kill-chain.html.

[36] "The VERIS Framework," https://verisframework.org/.

[37] MITRE, "CAPEC - Common Attack Pattern and Enumeration," https://capec.mitre.org/.

[38] B. E. Strom, A. Applebaum, D. P. Miller, K. C. Nickels, A. G. Pennington, and C. B. Thomas, "Mitre att&ck: Design and philosophy," in *Technical report*. The MITRE Corporation, 2018.

[39] "tactics, techniques, and procedures (ttp)," https://csrc.nist.gov/glossary/term/tactics_techniques_and_procedures, 2022.

[40] "What are tactics, techniques, and procedures (ttps)?" https://www.feroot.com/education-center/what-are-tactics-techniques-and-procedures-ttps/, 2022.

[41] "MITRE ATT&CK," https://attack.mitre.org, 2022.

[42] B. Strom, A. Applebaum, D. Miller, K. Nickels, A. Pennington, and C. Thomas, "Mitre att&ck: Design and philosophy," https://www.mitre.org/publications/technical-papers/mitre-attack-design-and-philosophy, MITRE, Tech. Rep., 2020.

[43] "Input capture T1056 - Enterprise — MITRE ATT&CK," https://attack.mitre.org/techniques/T1056/, 2022.

[44] "Flawedammyy — software 0381 — mitre att&ck," https://attack.mitre.org/software/S0381/, 2022.

[45] "IcedID Macro Ends in Nokoyawa Ransomware," https://thedfirreport.com/2023/05/22/icedid-macro-ends-in-nokoyawa-ransomware/, 2014.

[46] "Collect, Exfiltrate, Sleep, Repeat," https://thedfirreport.com/2023/02/06/collect-exfiltrate-sleep-repeat/, 2014.

[47] M. Grootendorst, "Creating a class-based tf-idf with scikit-learn," https://towardsdatascience.com/creating-a-class-based-tf-idf-with-scikit-learn-caea7b15b858.

[48] M. L. McHugh, "Interrater reliability: the kappa statistic," *Biochemia Medica*, pp. 276–282, 2012.

[49] J. Opitz and S. Burst, "Macro f1 and macro f1," *arXiv preprint arXiv:1911.03347*, 2019.

[50] J. Pustejovsky, J. M. Castano, R. Ingria, R. Sauri, R. J. Gaizauskas, A. Setzer, G. Katz, and D. R. Radev, "Timeml: Robust specification of event and temporal expressions in text." *New directions in question answering*, vol. 3, pp. 28–34, 2003.

[51] "TimeML Markup Language for Temporal and Event Expressions," https://timeml.github.io/site/publications/timeMLdocs/annguide_1.2.1.pdf.

[52] L. R. Derczynski, *Automatically ordering events and times in text*. Springer, 2017, p. 80.

[53] R. J. Howarth, "Dictionary of mathematical geosciences," *Cham: Springer*, 2017.

[54] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. Pearson, 2023, draft of Jan 7, 2023.

[55] Doron Karmi, "A Look Into Konni 2019 Campaign," https://medium.com/d-hunter/a-look-into-konni-2019-campaign-b45a0f321e9b, 2020.

[56] G. Piatetsky-Shapiro, "Discovery, analysis, and presentation of strong rules," *Knowledge Discovery in Data-bases*, pp. 229–248, 1991.

[57] M. Hahsler, "A probabilistic comparison of commonly used interest measures for association rules," https://mhahsler.github.io/arules/docs/measures.pdf.

[58] S. Harris and D. Harris, *Digital design and computer architecture*. Morgan Kaufmann, 2015.

[59] S. Ö. Arik and T. Pfister, "Tabnet: Attentive interpretable tabular learning," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 35, no. 8, 2021, pp. 6679–6687.

[60] M. Schlichtkrull, T. N. Kipf, P. Bloem, R. Van Den Berg, I. Titov, and M. Welling, "Modeling relational data with graph convolutional networks," in *The Semantic Web: 15th International Conference, ESWC 2018, Heraklion, Crete, Greece, June 3–7, 2018, Proceedings 15*. Springer, 2018, pp. 593–607.

[61] N. Chinchor and B. M. Sundheim, "Muc-5 evaluation metrics," in *Fifth Message Understanding Conference (MUC-5): Proceedings of a Conference Held in Baltimore, Maryland, August 25-27, 1993*, 1993.

[62] J. Zheng, W. W. Chapman, R. S. Crowley, and G. K. Savova, "Coreference resolution: A review of general methodologies and applications in the clinical domain," *Journal of biomedical informatics*, vol. 44, no. 6, pp. 1113–1122, 2011.

[63] Y. Wang, H. Tong, Z. Zhu, and Y. Li, "Nested named entity recognition: a survey," *ACM Transactions on Knowledge Discovery from Data (TKDD)*, vol. 16, no. 6, pp. 1–29, 2022.

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